THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 81	4+(12-1)x7
2	**(c)** 18	Solve $a+(n-1)d=$ the given number.
3	**(a)** 156	Sum of first n even numbers is $n(n+1)$.
4	**(c)** 210	Add 20.
5	**(a)** 21	Sticks $=3+2(n-1)=2n+1$.
6	**(d)** 625	Consecutive triangular numbers add to a square.
7	**(d)** 128	Each term is double the one before it.
8	**(c)** 93	9+(15-1)x6
9	**(b)** 25	Solve $a+(n-1)d=$ the given number.
10	**(c)** 240	Sum of first n even numbers is $n(n+1)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Formula $a+(n-1)d=5+(n-1)\times 3=3n+2$ (1); 20th term $=62$ (1).	1+1
12	Terms are 3, 7, ... each 4 more than the last (1); 35 is a term but 36 lies between two terms, so it is not a term (1).	1+1
13	Odd: $30^2=900$ (1); Even: $30\times 31=930$ (1).	1+1
14	Formula $a+(n-1)d=2+(n-1)\times 7=7n+-5$ (1); 15th term $=100$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sticks for 12 triangles $=3+2(12-1)=25$ (1); For 35 sticks: $2n+1=35$ so $n=16$ (1); 16 triangles (1).	1+1+1
16	There are 30 odd numbers, sum $=900$ (1); Even numbers: sum $=30\times 31$ (1); $=930$ (1).	1+1+1
17	$T_{20}=210$, $T_{21}=231$ (1); Sum $=441$ (1); $21^2=441$ matches (1).	1+1+1
	OR: Sticks for 15 triangles $=3+2(15-1)=31$ (1); For 41 sticks: $2n+1=41$ so $n=19$ (1); 19 triangles (1).	1+1+1
18	There are 40 odd numbers, sum $=1600$ (1); Even numbers: sum $=40\times 41$ (1); $=1640$ (1).	1+1+1
	OR: $T_{24}=300$, $T_{25}=325$ (1); Sum $=625$ (1); $25^2=625$ matches (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	10th figure $=5+(10-1)\times 4=41$ (1); More than 91 needs $(91-5)\div 4$ more steps (1); Steps needed: 13 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 9 numbers $=\frac{9\times 10}{2}=45$ (1); Sum of cubes $=2025$ (1); $45^2=2025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
20	12th figure $=3+(12-1)\times 5=58$ (1); More than 108 needs $(108-3)\div 5$ more steps (1); Steps needed: 11 (1); Figure number 23 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 10 numbers $=\frac{10\times 11}{2}=55$ (1); Sum of cubes $=3025$ (1); $55^2=3025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 300	Add 24.
2	**(a)** 31	Sticks $= 3 + 2(n-1) = 2n + 1$.
3	**(a)** 900	Consecutive triangular numbers add to a square.
4	**(c)** 256	Each term is double the one before it.
5	**(c)** 93	13+(11-1)x8
6	**(b)** 13	Solve $a + (n-1)d =$ the given number.
7	**(c)** 342	Sum of first n even numbers is $n(n+1)$.
8	**(c)** 406	Add 28.
9	**(b)** 41	Sticks $= 3 + 2(n-1) = 2n + 1$.
10	**(d)** 1225	Consecutive triangular numbers add to a square.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Terms are 6, 11, ... each 5 more than the last (1); 61 is a term but 62 lies between two terms, so it is not a term (1).	1+1
12	Odd: $35^2 = 1225$ (1); Even: $35 \times 36 = 1260$ (1).	1+1
13	Formula $a + (n-1)d = 11 + (n-1) \times 4 = 4n + 7$ (1); 18th term $= 79$ (1).	1+1
14	Terms are 2, 8, ... each 6 more than the last (1); 80 is a term but 81 lies between two terms, so it is not a term (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sticks for 18 triangles $= 3 + 2(18-1) = 37$ (1); For 47 sticks: $2n + 1 = 47$ so $n = 22$ (1); 22 triangles (1).	1+1+1
16	There are 45 odd numbers, sum $= 2025$ (1); Even numbers: sum $= 45 \times 46$ (1); $= 2070$ (1).	1+1+1
17	$T_{28} = 406$, $T_{29} = 435$ (1); Sum $= 841$ (1); $29^2 = 841$ matches (1).	1+1+1
	OR: Sticks for 21 triangles $= 3 + 2(21-1) = 43$ (1); For 53 sticks: $2n + 1 = 53$ so $n = 25$ (1); 25 triangles (1).	1+1+1
18	There are 50 odd numbers, sum $= 2500$ (1); Even numbers: sum $= 50 \times 51$ (1); $= 2550$ (1).	1+1+1
	OR: $T_{32} = 528$, $T_{33} = 561$ (1); Sum $= 1089$ (1); $33^2 = 1089$ matches (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	9th figure $= 8 + (9-1) \times 6 = 56$ (1); More than 106 needs $(106-8) \div 6$ more steps (1); Steps needed: 9 (1); Figure number 18 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 11 numbers $= \frac{11 \times 12}{2} = 66$ (1); Sum of cubes $= 4356$ (1); $66^2 = 4356$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
20	11th figure $= 4 + (11-1) \times 7 = 74$ (1); More than 124 needs $(124-4) \div 7$ more steps (1); Steps needed: 8 (1); Figure number 19 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 12 numbers $= \frac{12 \times 13}{2} = 78$ (1); Sum of cubes $= 6084$ (1); $78^2 = 6084$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 512	Each term is double the one before it.
2	**(d)** 122	5+(14-1)x9
3	**(a)** 22	Solve $a+(n-1)d=$ the given number.
4	**(a)** 506	Sum of first n even numbers is $n(n+1)$.
5	**(b)** 528	Add 32.
6	**(b)** 51	Sticks $=3+2(n-1)=2n+1$.
7	**(d)** 1600	Consecutive triangular numbers add to a square.
8	**(b)** 1024	Each term is double the one before it.
9	**(c)** 104	16+(9-1)x11
10	**(b)** 31	Solve $a+(n-1)d=$ the given number.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Odd: $40^2=1600$ (1); Even: $40\times41=1640$ (1).	1+1
12	Formula $a+(n-1)d=6+(n-1)\times5=5n+1$ (1); 25th term $=126$ (1).	1+1
13	Terms are 9, 17, ... each 8 more than the last (1); 81 is a term but 82 lies between two terms, so it is not a term (1).	1+1
14	Odd: $50^2=2500$ (1); Even: $50\times51=2550$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sticks for 24 triangles $=3+2(24-1)=49$ (1); For 59 sticks: $2n+1=59$ so $n=28$ (1); 28 triangles (1).	1+1+1
16	There are 60 odd numbers, sum $=3600$ (1); Even numbers: sum $=60\times61$ (1); $=3660$ (1).	1+1+1
17	$T_{36}=666$, $T_{37}=703$ (1); Sum $=1369$ (1); $37^2=1369$ matches (1).	1+1+1
	OR: Sticks for 27 triangles $=3+2(27-1)=55$ (1); For 65 sticks: $2n+1=65$ so $n=31$ (1); 31 triangles (1).	1+1+1
18	There are 75 odd numbers, sum $=5625$ (1); Even numbers: sum $=75\times76$ (1); $=5700$ (1).	1+1+1
	OR: $T_{40}=820$, $T_{41}=861$ (1); Sum $=1681$ (1); $41^2=1681$ matches (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	14th figure $=6+(14-1)\times3=45$ (1); More than 95 needs $(95-6)\div3$ more steps (1); Steps needed: 17 (1); Figure number 31 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 13 numbers $=\frac{13\times14}{2}=91$ (1); Sum of cubes $=8281$ (1); $91^2=8281$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2
20	8th figure $=2+(8-1)\times8=58$ (1); More than 108 needs $(108-2)\div8$ more steps (1); Steps needed: 7 (1); Figure number 15 (1); Check by substituting (1).	1+1+1+1+1
	OR: Sum of first 14 numbers $=\frac{14\times15}{2}=105$ (1); Sum of cubes $=11025$ (1); $105^2=11025$ (1); They are equal, so the cube sum is a perfect square (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.